Gu Shiyuan

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https://github.com/1MGSY0 | Design Portfolio Canva link**: https://tinyurl.com/42yejn7b**

EDUCATION

Nanyang Technological University, Singapore Aug 2022 – May 2026

- Bachelor of Engineering (Information Engineering and Media) with Major in Data Analytics

Hwa Chong Junior Collage, Singapore Jan 2020 – Dec 2022

- Singapore-Cambridge GCE Advanced Level

Crescent Girls' School Jan 2015 – Jun 2019

- Singapore-Cambridge GCE Ordinary Level

CERTIFICATE

Coursera May 2025 – Dec 2025

- Cybersecurity Specialization by Google
- Project Management Specialization by Google
- Business Intelligence Specialization by Google
- AI Essentials by Google

Japanese Language Proficiency Test N1 Aug 2014

ACADEMIC PROJECTS

TRPG Game Engine Development Project Mar 2025 – Present

(Demo: https://youtu.be/JAKOZiRdv_o)

- Co-developed a modular TRPG game engine in C++.
- Architected entity and implemented core subsystems (rendering with bgfx, asset management, dialogue manager)
- Designed flowchart-based scenario builder with Draw.io + Figma mockups.

Interactive System Design – User-focused Unity VR game development Jan 2025 – Mar 2025

(Demo: https://youtu.be/v1vsfF0JHzE)

- Co-Developed a VR audio maze game using spatial audio and AI voice with Unity and deployed on Meta Headset
- Conducted user testing prototypes and focus groups, increasing replay interest by 40% after iterative design improvements
- Applied human-centered design principles, accessibility heuristics, and interaction design methodologies to ensure inclusivity for visually impaired users

Predictive Analytics – House Price Prediction Jan 2025 – Mar 2025

- Engineered predictive models (Lasso Regression, XGBoost) using Python to forecast house prices with 92% R^2 test accuracy, improving RMSE by 72% over baseline.
- Applied advanced feature engineering, log transformations, and model tuning (GridSearchCV) on 70+ variables from the Ames dataset.
- Built a reproducible machine learning pipeline with scikit-learn, pandas, and visualized performance metrics for stakeholder-ready insights

Web Programming Project – Design your own e-shop website Mar 2024 – Mar 2024

(Demo: https://youtu.be/v1vsfF0JHzE)

- Design a handcraft e-commers website prototype on Figma.
- Create Web Application based on the Figma prototype with SQL, HTML, CSS, JSP and Java servlets and deployed locally with Tomcat with a 2-week timeframe.

Unity Engine Game Project – Design a 3D original game Feb 2024 – Feb 2024

(Demo: https://youtu.be/LgxkFNPfUVc)

- Plan and creatively design a 3D original game.
- Implement it on Unity with C# scripts and presented it within 2-Week development timeframe.

WORK EXPERIENCE

Centific, Singapore Nov 2025 – Present
AI Data Associate

- Annotate Japanese and Chinese content based on given standards and guidelines.
- Conduct research on various topics using reliable internet sources.
- Provide timely and accurate evaluation.

HTX (Home Team Science & Technology Agency), Singapore Jul 2024 – Dec 2024
Intelligent Video Analytics Intern

- Developed a AI role-play training platform using React frontend and Hugging Face Transformers for backend NLP and contextual grading
- Containerized full-stack deployment with Docker for seamless integration and future delivery to stakeholders

Gecipe E-Sports English, Japan
Quality Manager (Sub-Head Coach) Aug 2023 – Dec 2024

- Supervising the lesson quality of over 100 coaches in the quality assurance team and contributing to record high user rating of over 4.78/5 in February 2024.
- Collaborated with the team to design lesson plans and implement new initiatives, resulting in savings of JYP 500K
- Delivered trainings on communication, and English to Japanese speaking students

LEADERSHIP & CO-CURRICULAR ACTIVITIES

NTU Tennis Varsity Jun 2024 – Present

- Represent NTU to play competitive tennis in Singapore University Games

Visual Arts Society Jun 2023 – Sep 2024
Honorary General Secretary

- Set goals and targets to achieve with the other committee, successfully increased intake of 100 new members
- Ensured compliance with PDPA while collecting data and developed a new communication system within the club

NTU Tennis Club Jun 2023 – Present
Financial Controller

- Oversee and conduct event management and financial spending of the club of 120 members
- Lead the team of EXCO in club activity by setting goals and the vision of the club
- Automated member registration and selection process
- Developed Telegram Bot to track shed usage and deployed on Google cloud

Lead of Publicity and Publications

- Oversaw publicity efforts on Instagram and developed outreach strategies to deliver consistent message across various platforms reaching a breakthrough 20% increase of new club members
- Lead the team of designers to complete design and produce projects on time
- Designed Social media posts, T-shirt designs and Logo

SKILLS & INTERESTS

Languages: Native Proficiency in Japanese, English and Chinese
Technical Skills:
- Programming: Python, Java, C#, C++, HTML, JSP, CSS, JavaScript, SQL, React
- Frameworks & Tools: Git, GitHub, Docker, Unity, Android Studio, Blender, Eclipse

Applications: MATLAB, Microsoft Office, Power Automate, Figma, DaVinci Resolve, Clip Studio Paint, Procreate, Autodesk Inventor, Google cloud services

Interests: Tennis, Digital Art, J-Rock Band Singer/Bassist